

A gloved hand is holding a conical flask containing a red liquid. A burette is positioned above the flask, dispensing a clear liquid into it. The setup is on a white base.

Analysis of Antacid By Titration

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Titration:

Titration is a common laboratory method of quantitative chemical analysis and is a technique where a solution of known concentration is used to determine the unknown concentration of a solution.

The addition of standard solution (known concentration) (titrant) from the burette to a known volume of solution (unknown concentration) that will react with the titrant in the presence of indicator.

The addition is continued until the color is changed the reaction is complete, and measuring the titrant volume so as to know the concentration of the unknown solution.

An indicator is used to detect the end of the reaction, the endpoint.

An indicator: is a chemical substance that changes color due to changes in pH.

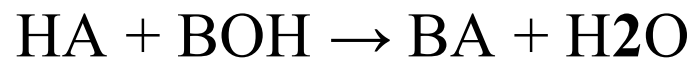
End point: the point during a titration when an indicator shows that the amount of reactant necessary for a complete reaction has been added to a solution.

Titration can be classified as:

1. Acid-base titrations or acidimetry and alkalimetry
2. Oxidation-reduction titrations or redox titrations
3. Precipitation titrations
4. Complexometric titrations.

1. Acid-base titrations :

Both acid and base titrations involve neutralization. In these titrations H^+ ions of the acid combine with OH^- ions of the alkali to form unionized molecules of water.



Acid Alkali Salt Water



Phenolphthalein is a suitable indicator in the titrations of strong alkalis against strong or weak acids.

Methyl orange is used as an indicator in the titrations of strong acids against strong and weak alkalis.

2. Oxidation-reduction titrations :

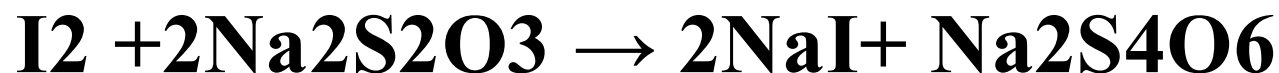
The titrations are based on oxidation-reduction reaction between an oxidizing agent and a reducing agent are called redox titrations.

The chemical reaction proceed with transfer of electrons (loss or gain of electrons) among the reacting ions in aqueous solutions.

- The reduction of free iodine to iodide ions and
- oxidation of iodide ions to free occurs in these titrations.

- $\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$ (reduction)
- $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$ (oxidation)

- Free iodine is titrated against a standard reducing agent usually with sodium thiosulphate. can be estimate by this method.

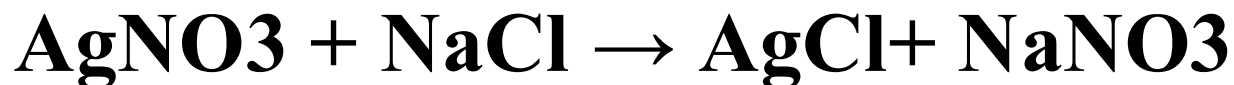


- In iodometric titrations, Starch is used as an indicator solution gives blue or violet color with free iodine. At the end point the blue or violet color disappears when iodine is completely change to iodide.

3. Precipitation titration:

Is a type of titration which involves the formation of precipitate during the titration technique.

For example, when a solution of silver nitrate is added to a solution of sodium chloride , a white precipitate of silver chloride or silver thiocyanate is formed.



4. Complexometric titrations:

Titration relies on the formation of a complex between the analyte and the titrant.

EDTA (ethylene diamine tetra-acetic acid) is a useful reagent which forms complexes with metals. In the form of disodium salt, it is used to estimate Ca^{2+} and Mg^{2+} ions in the presence of eriochrome black-T as an indicator.

Analysis of anti-acid By titration

The natural environment of our stomach is quite acidic, which is mostly hydrochloric acid has a pH of 1.0. The acidic environment created by hydrochloric acid converts the pepsinogen into its active form, pepsin. Pepsin is an enzymes helps in the digestion of proteins by breaking down polypeptides into smaller peptides.

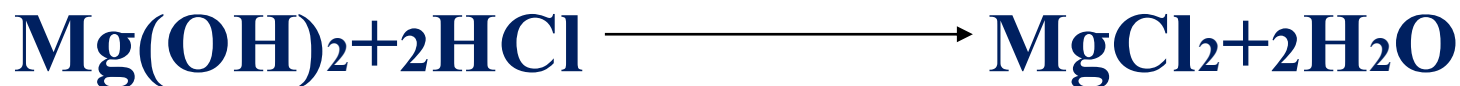
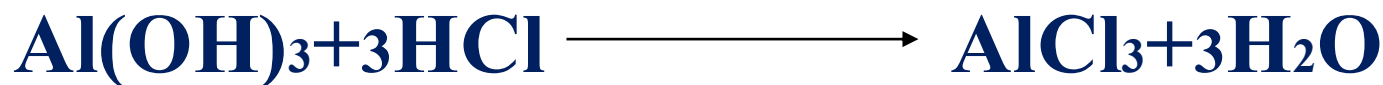
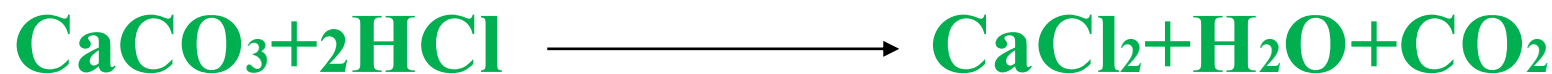
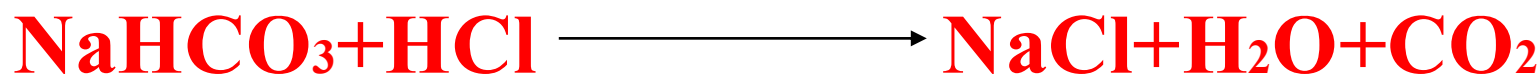
But too much acid in the stomach is not good either in the absence of food, or the strong acid, HCL, denatures the proteins in the stomach wall itself. If this goes on unchecked, it may cause stomach or duodenal ulcers. we feel the excess acidity in our stomach. Such sensations are called “heartburn” or “sour stomach” to relieve it we take antacids in tablet or liquid form.

Active ingredient of some antacid drugs

1- sodium bicarbonate

2- calcium carbonate

3- aluminum hydroxide and magnesium hydroxide



Procedure

- 1- Fill the burette with the standard (0.1N) HCl solution.
- 2- dissolve one tablet antacid in 100 ml distil water.
- 3- pipette out 10ml of the (antacid tablet solution) into the
conical flask

- 3- Add 3 drop of phenolphthalein as indicator
- 4- Titrate the antacid solution with HCl until the color changes at the end point from pink to color less, repeat the titration to ensure the end point and measure the volume of HCl (V_1) which is required to neutralize the total alkalinity.

Calculations:-

$$(N_a \times V_a) \text{ HCl} = (N_b \times V_b) \text{ Antacid}$$

$$0.1 \times V_1 = 10 \times N \text{ antacid}$$

$$N \text{ antacid} = ?$$